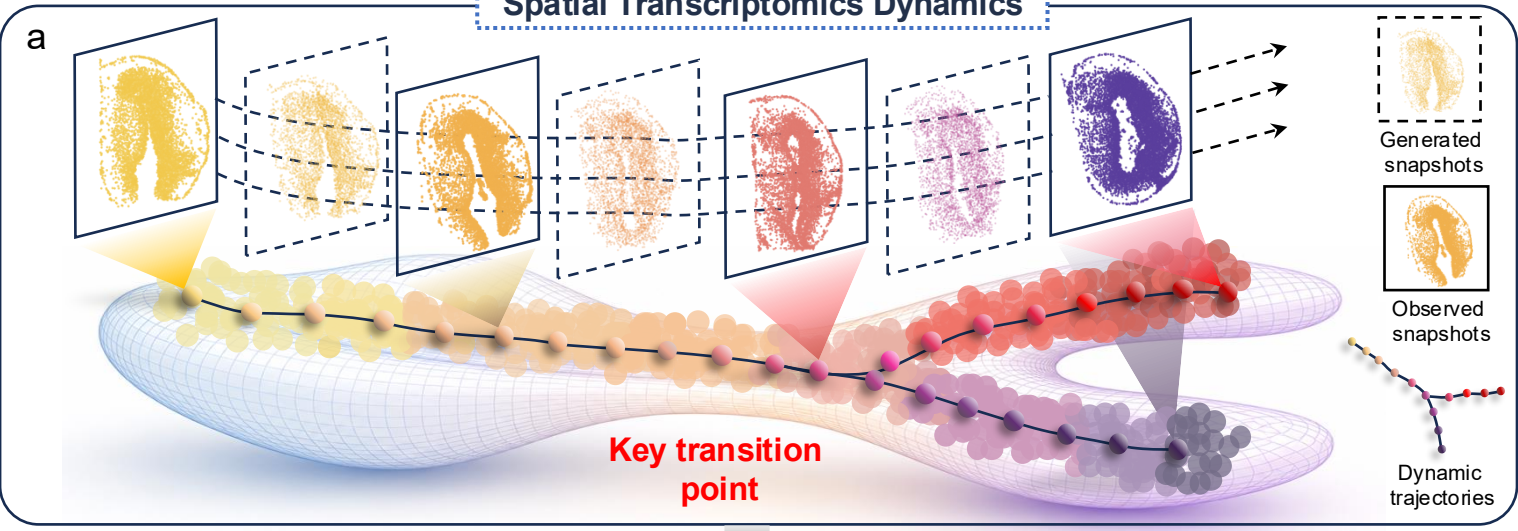
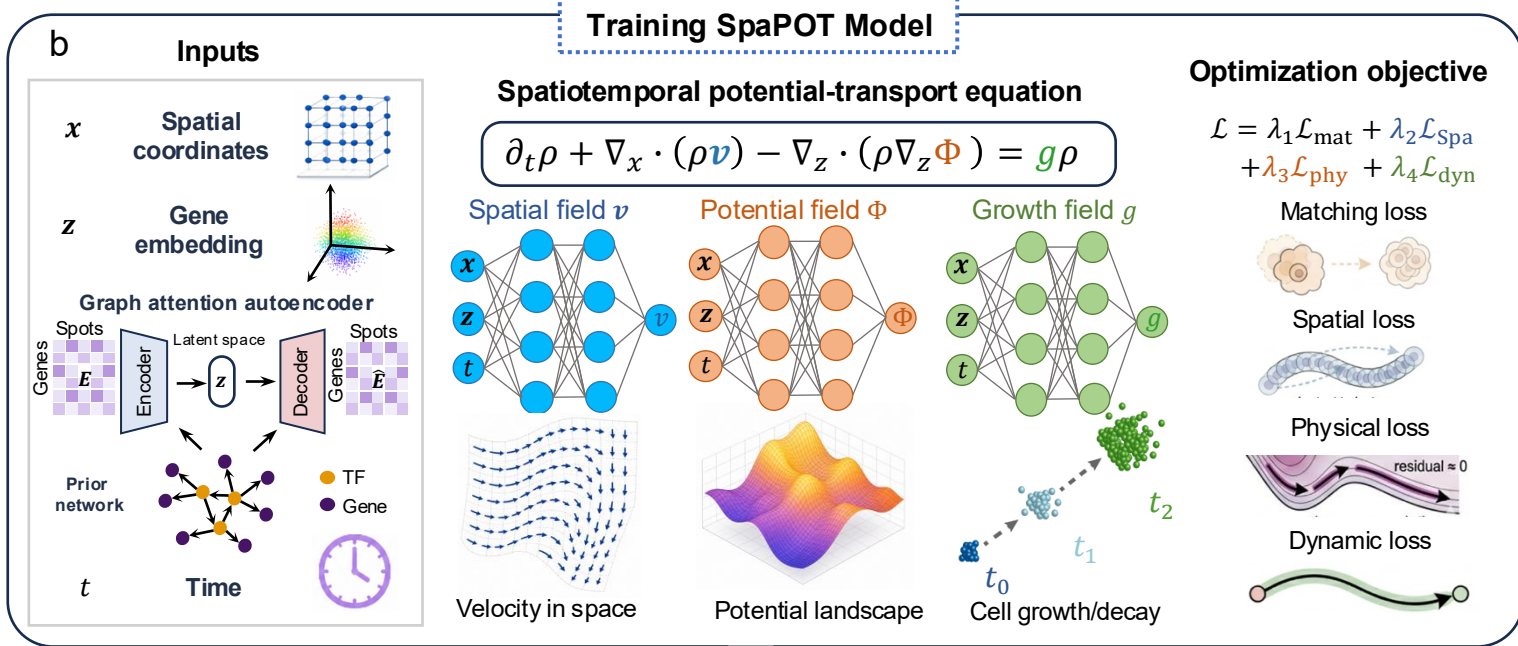


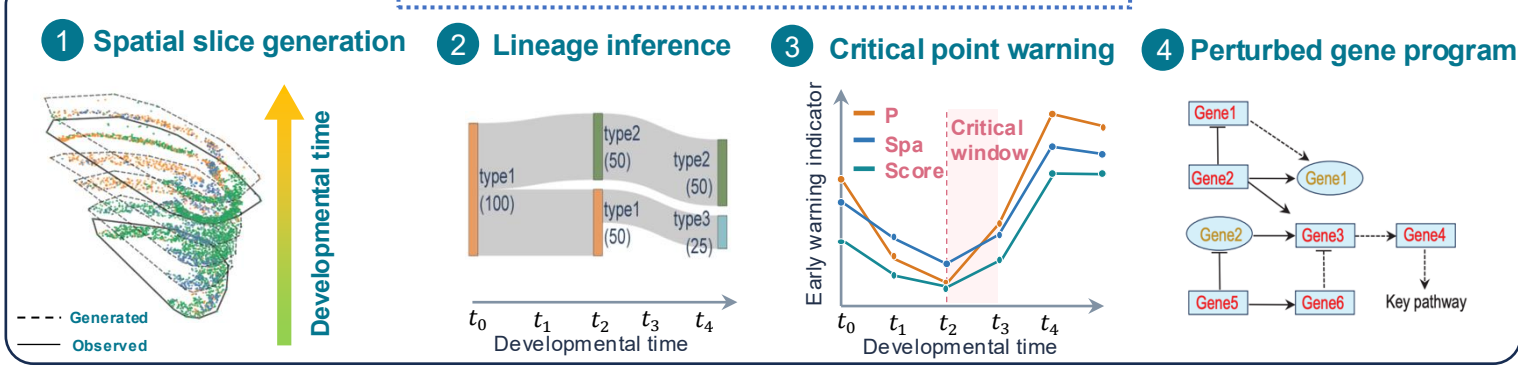
Spatial Transcriptomics Dynamics



Training SpaPOT Model

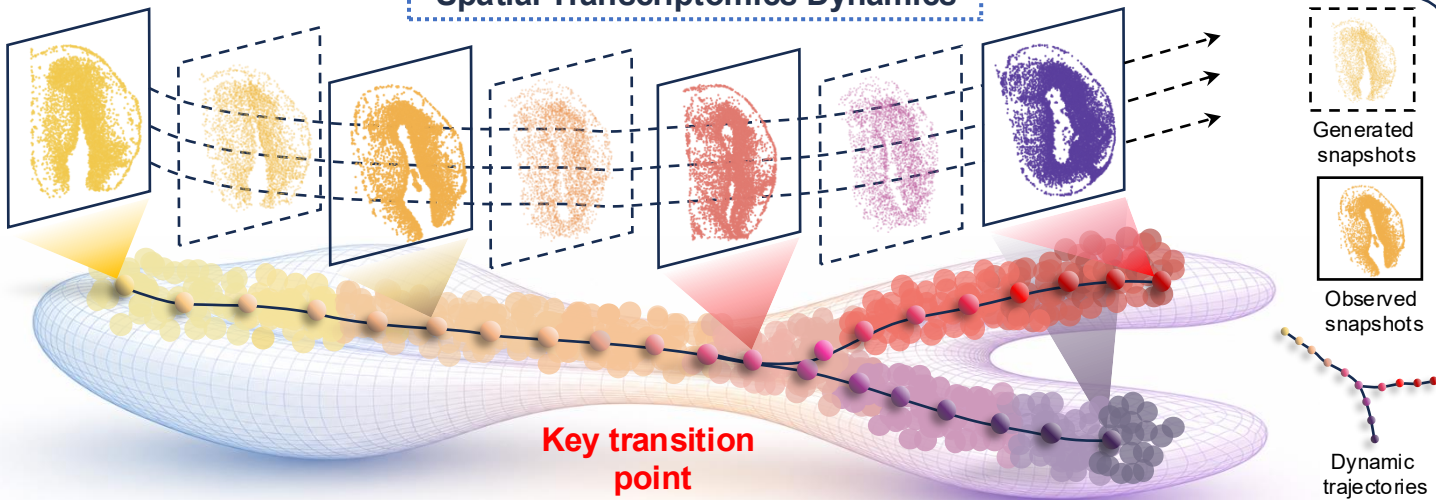


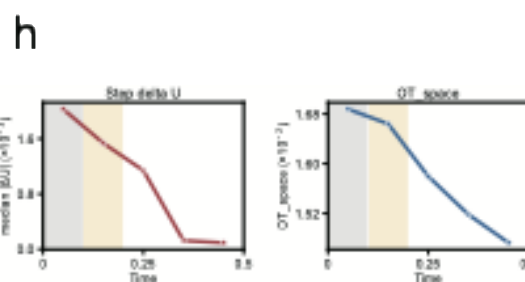
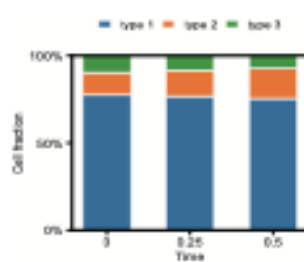
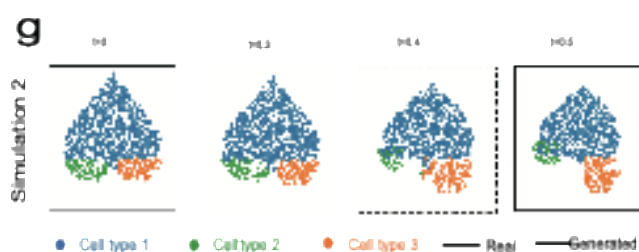
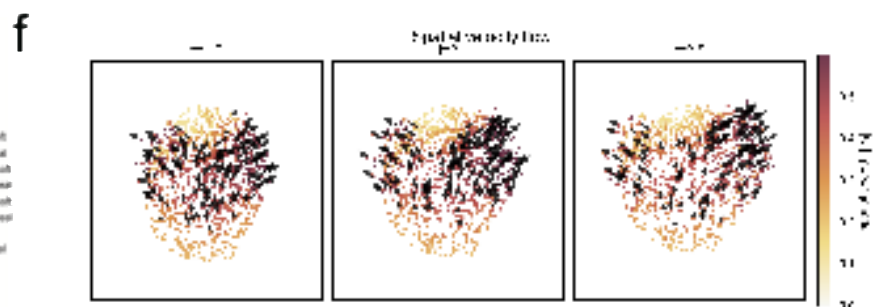
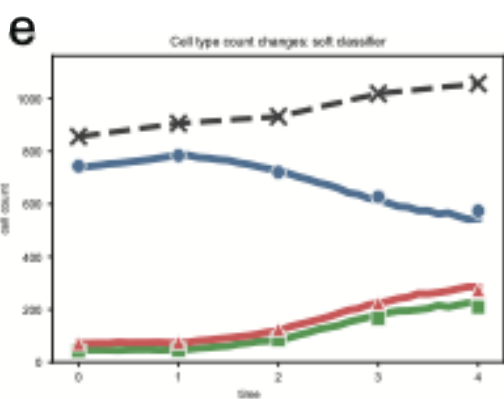
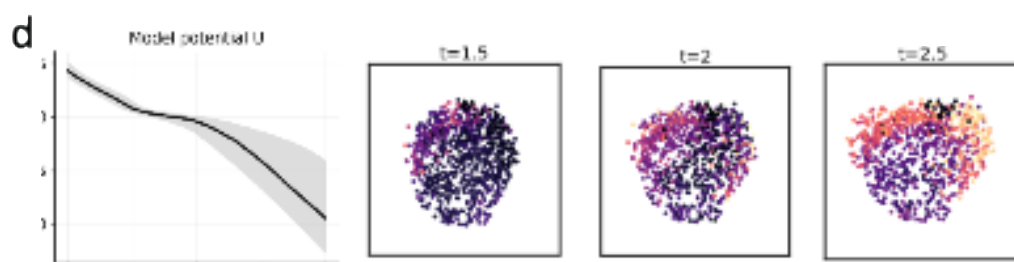
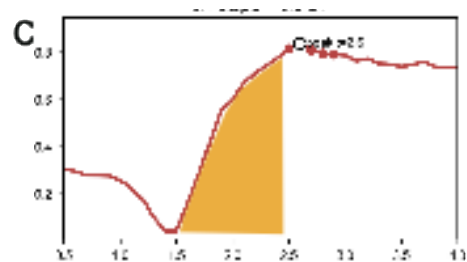
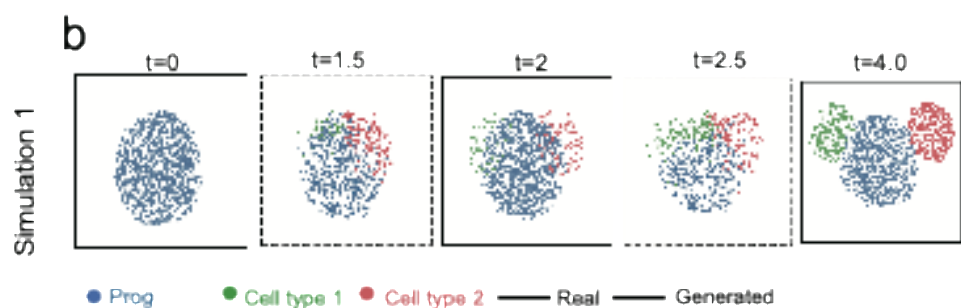
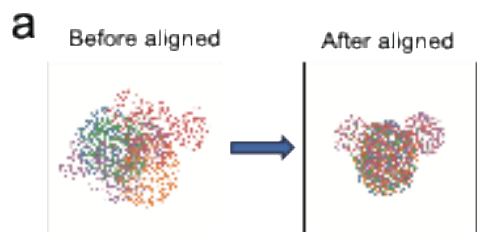
Downstream Analyses and Applications

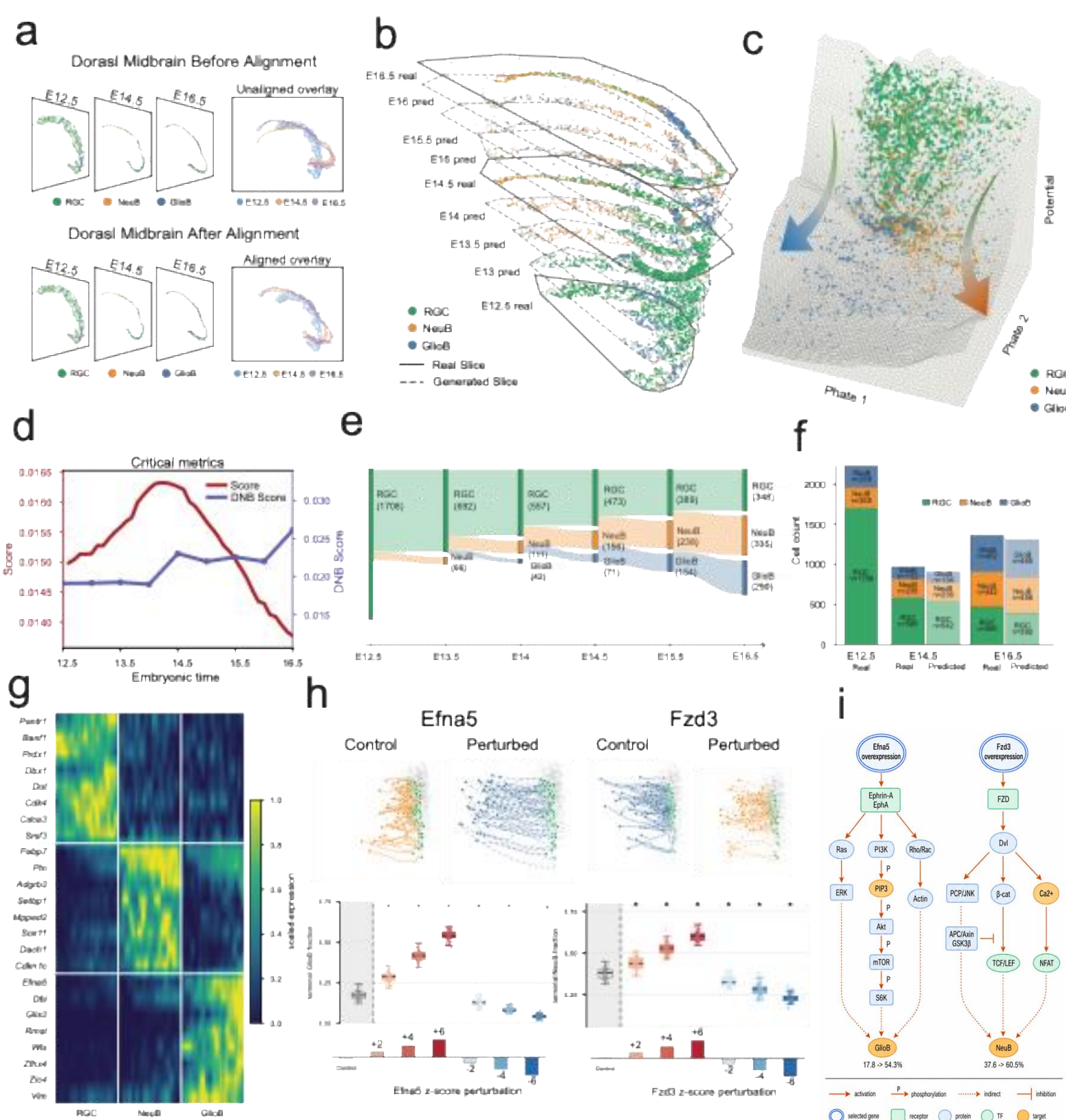


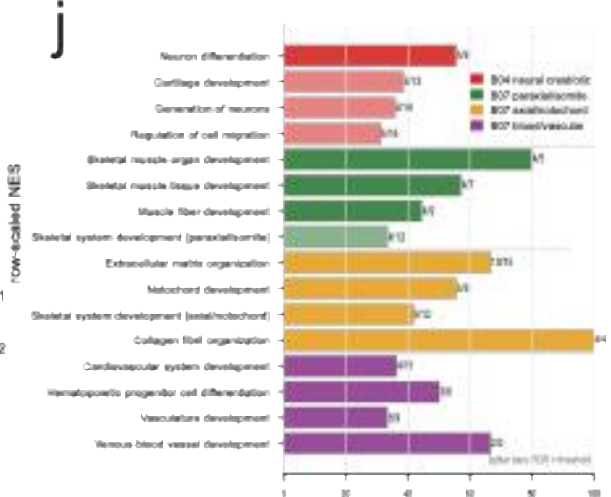
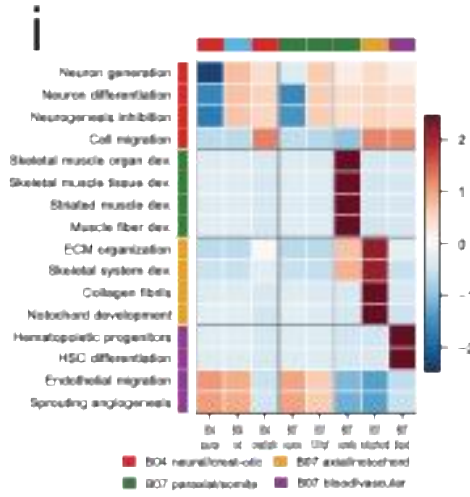
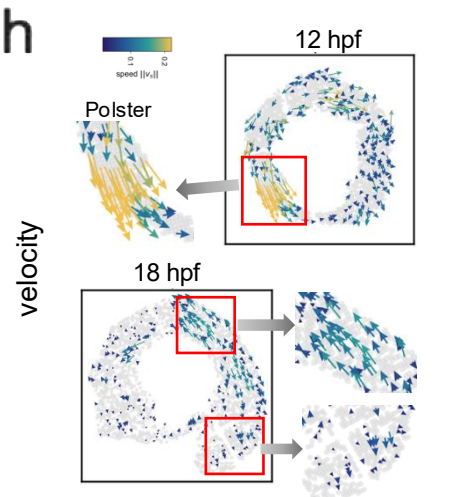
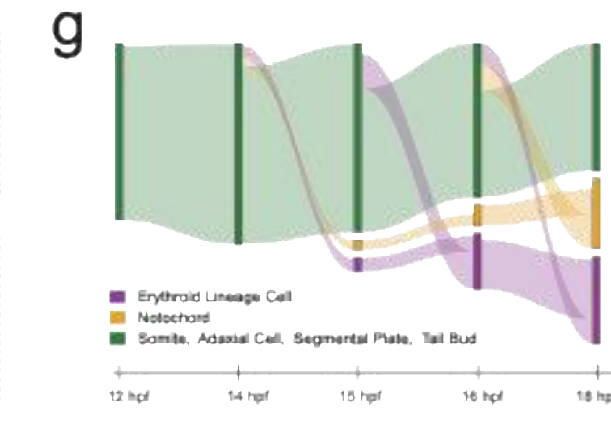
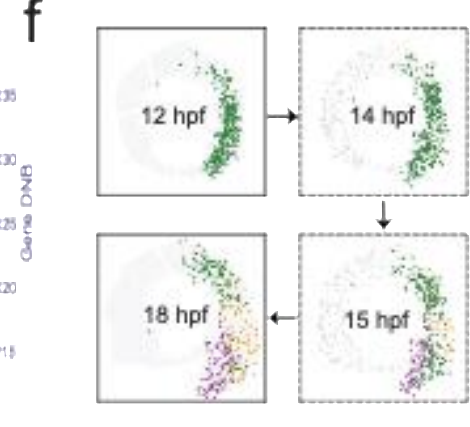
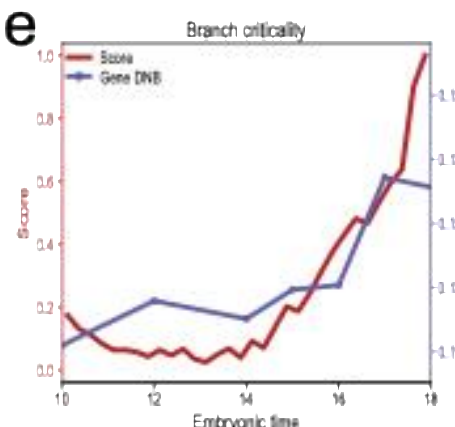
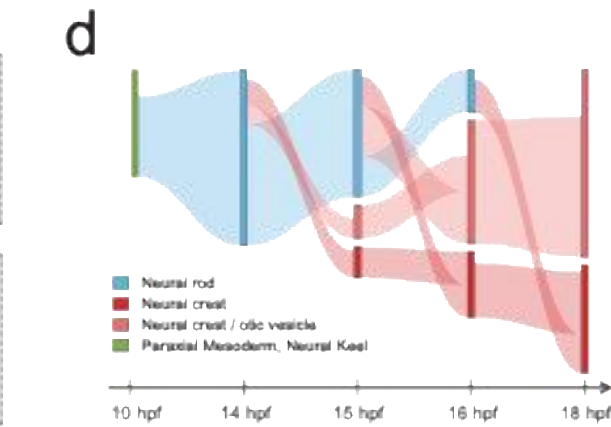
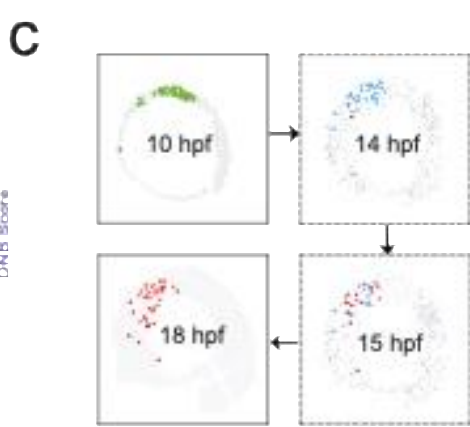
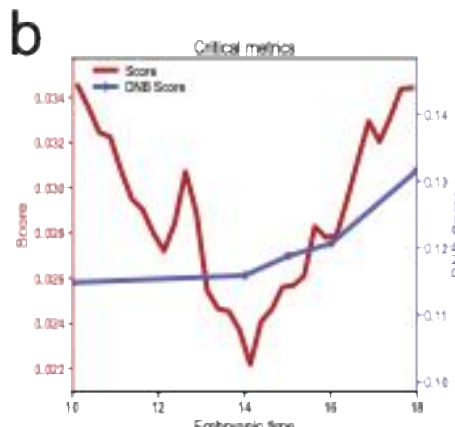
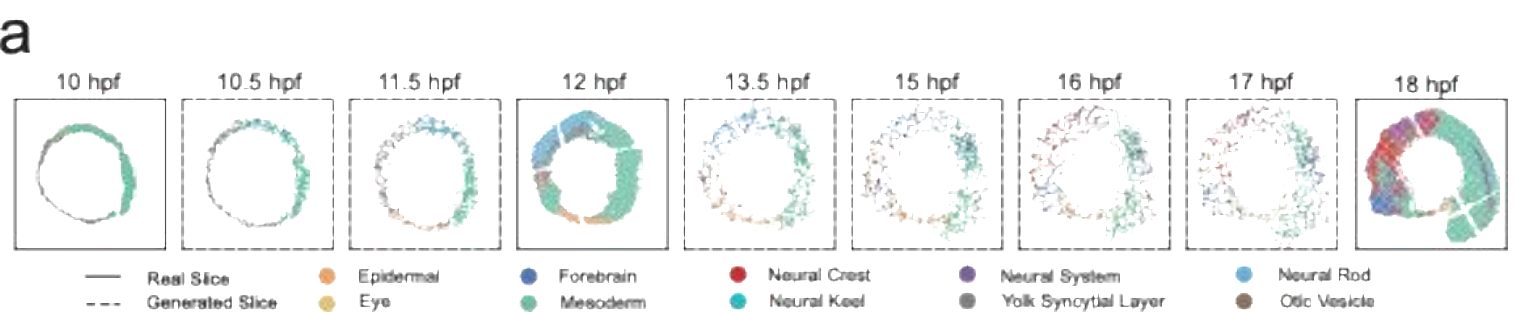
Spatial Transcriptomics Dynamics

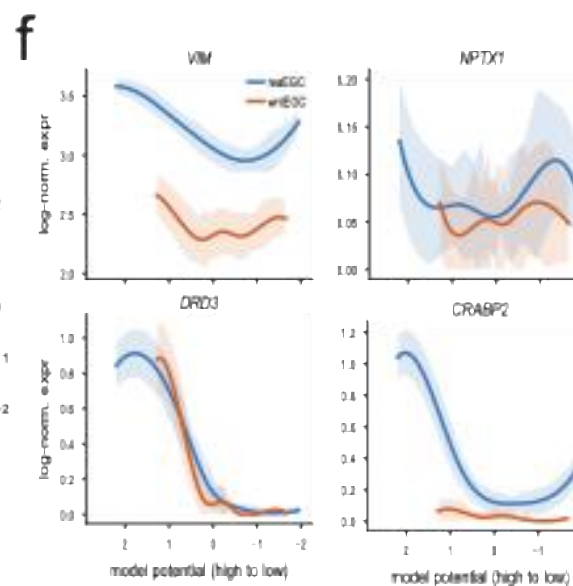
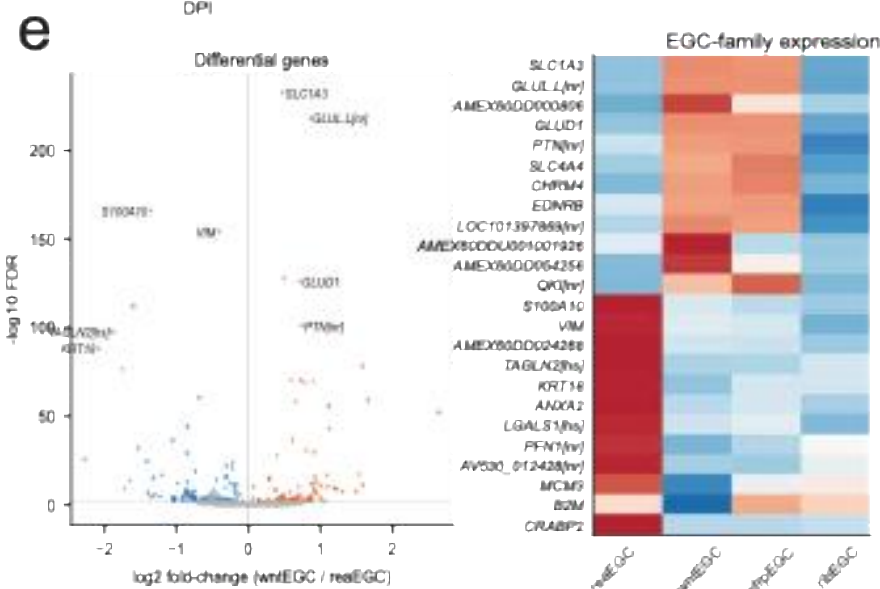
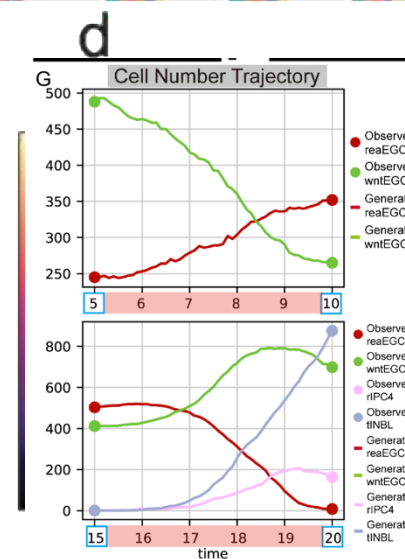
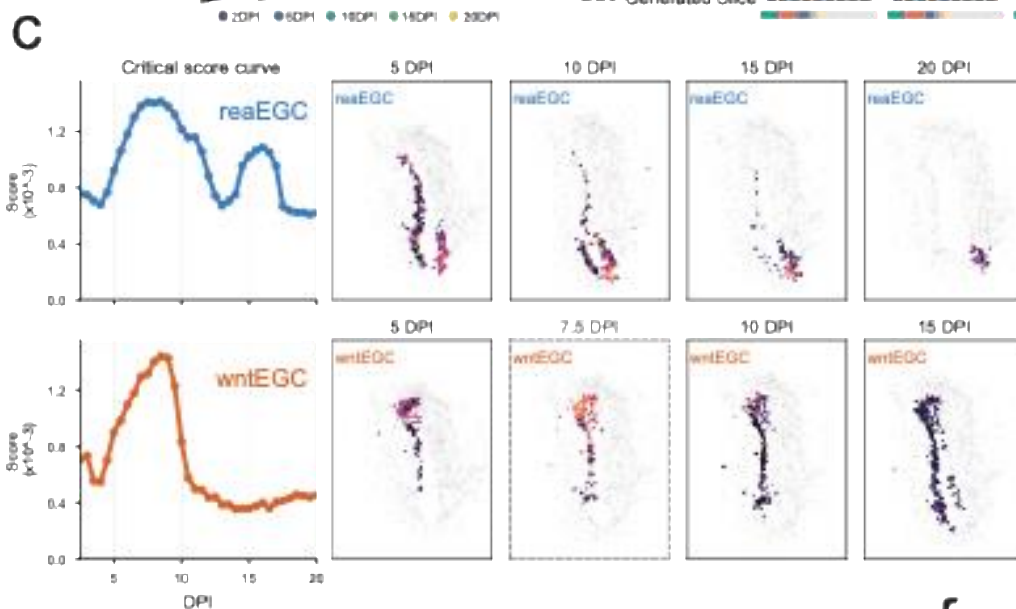
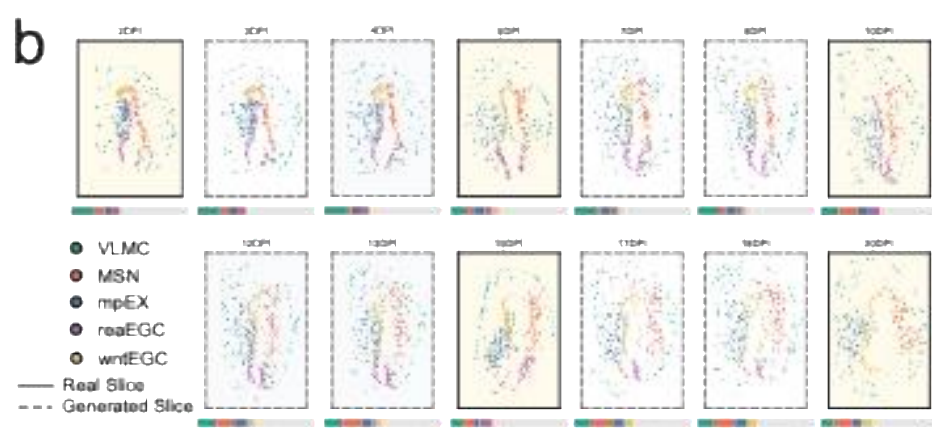
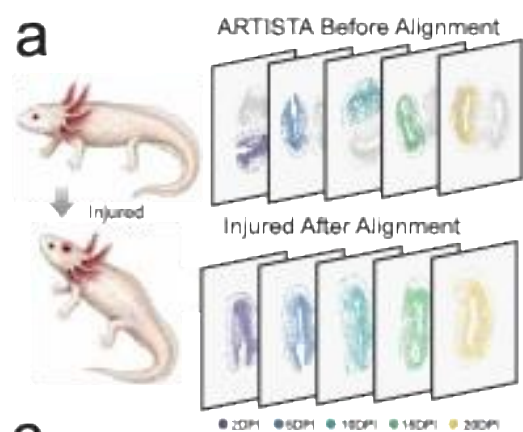
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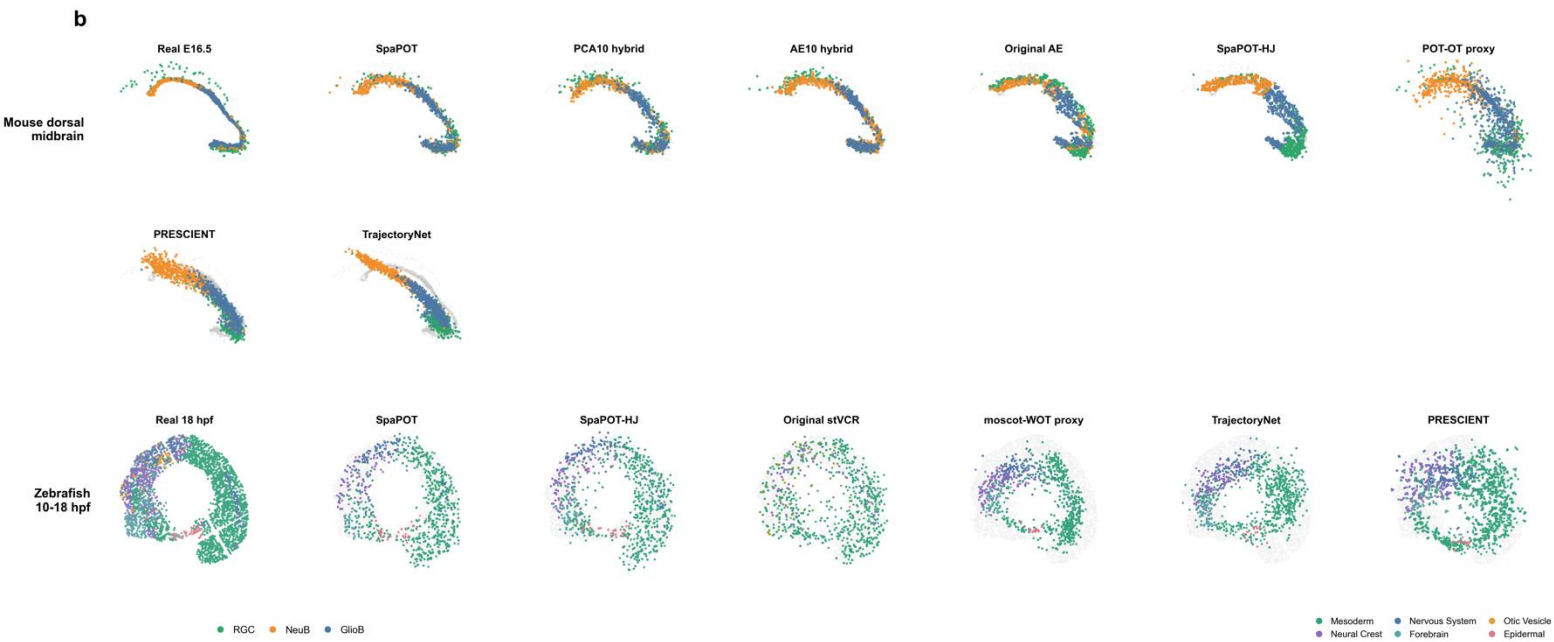












b

Mouse dorsal midbrain								Zebrafish 10-18 hpf							
Aggregate	0.92	0.87	0.77	0.61	0.52	0.37	0.23	0.22	Aggregate	0.77	0.76	0.73	0.49	0.36	0.33
State L1									State L1						
Spatial Chamfer									Spatial Chamfer						
Spatial IoU									Spatial IoU						
	SpaPOT	PCA10 hybrid	POT-OT proxy	AE10 hybrid	Original AE	SpaPOT -HJ	PRESCIENT adapted	TrajectoryNet like		SpaPOT	SpaPOT -HJ	PRESCIENT adapted	TrajectoryNet like	Original stVCR	moscot-WOT proxy

Models	PAS TE											
Spatial alignment												
Spatial pseudotime												
Generative reconstruction												
State inference												
Fate allocation												
Mass dynamics												
State inference												
In silico perturb												
Critical window												



